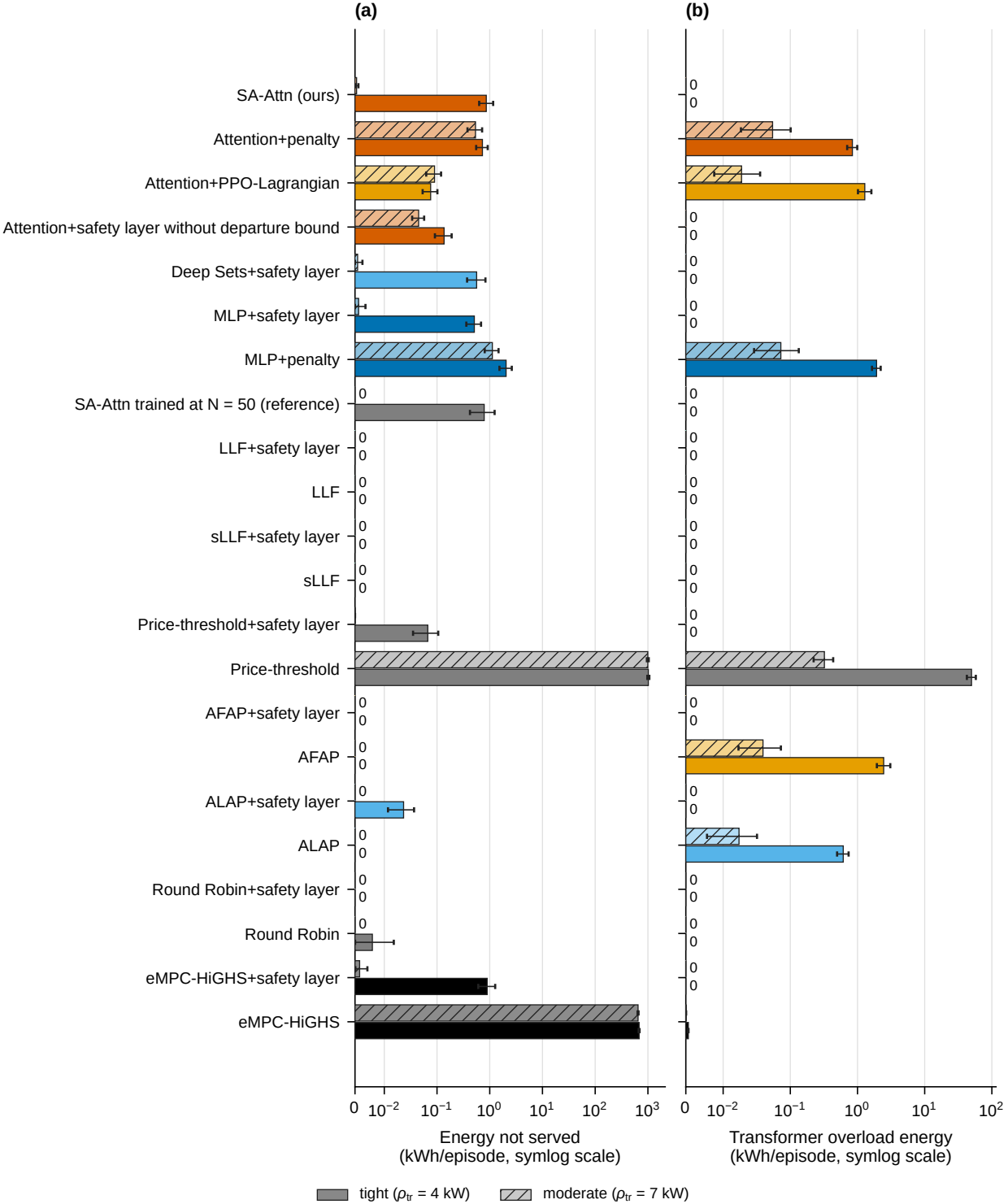
(a)
(b)
SA-Attn (ours)
Attention+penalty
Attention+PPO-Lagrangian
Attention+safety layer without departure bound
Deep Sets+safety layer
MLP+safety layer
MLP+penalty
SA-Attn trained at N = 50 (reference)
LLF+safety layer
LLF
sLLF+safety layer
sLLF
Price-threshold+safety layer
Price-threshold
AFAP+safety layer
AFAP
ALAP+safety layer
ALAP
Round Robin+safety layer
Round Robin
eMPC-HiGHS+safety layer
eMPC-HiGHS
0
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Energy not served
(kWh/episode, symlog scale)
Transformer overload energy
(kWh/episode, symlog scale)
tight (ρ_{tr} = 4 kW)
moderate (ρ_{tr} = 7 kW)